CHAPTER 1.

Theorem 1.10.

Let $K=(O,\mathcal{B})$, $K'=(O',\mathcal{B}')$ 2 frames on $\mathbb{A}^m$.

For any point $P \in \mathbb{A}^m$, we have:

$$[P]_{K'} = M_{B'B}\cdot([P]_K - [O']_K)$$

$$= M^{-1}_{BB'}\cdot([P]_K - [O']_K)$$

¿?

$$= M_{B'B}\cdot[P]_K + [O]_{K'} \quad (*)$$

Proof:

• The coordinates of a point are the components of its position vector:

$K' \to O'$
$K \to O$

$(*) \iff$

$$[\vec{O'P}]_{B'} = M_{B'B}\cdot([\vec{OP}]_B - [\vec{OO'}]_B)$$

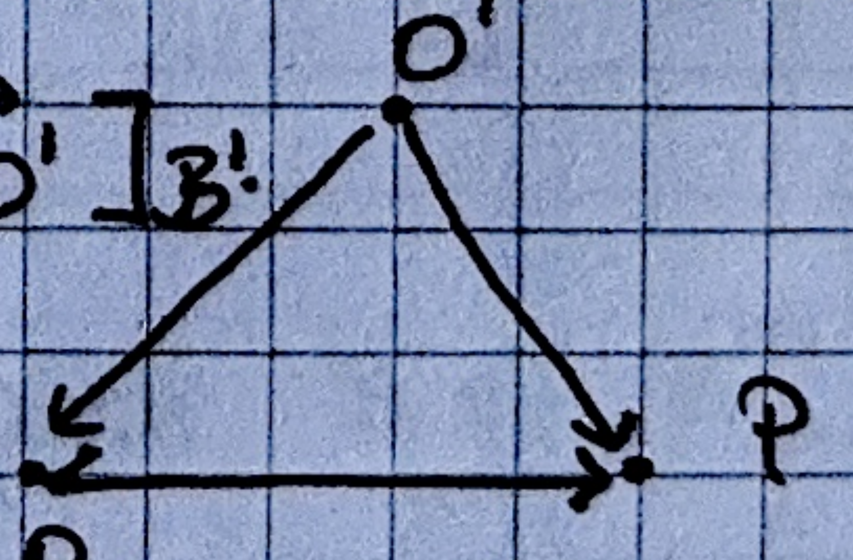
$$= M^{-1}_{BB'}\cdot([\vec{OP}]_B - [\vec{OO'}]_B)$$

$$= M_{B'B}\cdot[\vec{OP}]_B + [\vec{O'O}]_{B'}$$

$$-M_{B'B}\cdot[\vec{OO'}]_B = -[\vec{O'O}]_{B'} = [\vec{OO'}]_{B'}$$

$$[\vec{O'P}]_B = [\vec{OP}]_B - [\vec{OO'}]_B.$$

$$M_{B'B} = + M^{-1}_{BB'}$$



$\vec{O'O} + \vec{OP} = \vec{O'P}$.
$\vec{OP} - \vec{OO'} = \vec{O'P}$.

Expansion of Theorem 1.10.

$K = (O, B)$, $K' = (O', B')$ frames in $\mathbb{A}^n$.

- We know K' in terms of K:
- we know the coordinates of O' relative to K
- the components of $i_1', \dots, i_n'$ w.r. to B.
- $P \to$ its coordinates are relative to K

$\Rightarrow$ we can find $P \to$ coord. relative to K';

1) $M_{BB'} \to$ place $[i_1']_B, \dots, [i_n']_B$ on COLUMNS.

2) $M_{B'B} = M_{BB'}^{-1}$.

3) $[P]_{K'} = M_{B'B} \cdot ([P]_K - [O']_K)$

CHAPTER 2.

Theorem 2.6.

Frame $K = (O, B)$ in the affine space $\mathbb{A}^m$.

$$(S) \begin{cases} a_{11} \cdot x_1 + \dots + a_{1m} \cdot x_m = b_1 \\ \vdots \\ a_{t1} \cdot x_1 + \dots + a_{tm} \cdot x_m = b_t \end{cases}$$

→ system of linear equations in the unknowns $x_1, \dots, x_m$.

S is an affine space of dimension $d = m - r$. (rank ↑ r)

↳ the set of points $\in \mathbb{A}^m$ whose coordinates are solutions to (S)

$D(S)$ = direction space

↳ the vector subspace of $\mathbb{V}^m$ whose equations relative to B are given by:

$$D(S): \begin{cases} a_{11} \cdot x_1 + \dots + a_{1m} \cdot x_m = 0 \\ a_{t1} \cdot x_1 + \dots + a_{tm} \cdot x_m = 0 \end{cases}$$ homog. system.

? Conversely, for every affine subspace S of $\mathbb{A}^m$ of dimension d, $\exists$ system of $m-d$ lin. eq. in m unknowns whose sol. correspond to coord. of the points in S.

Proof.

"$\Longrightarrow$" S is an affine subspace of $\mathbb{A}^m$

$S = \{x_1, \dots, x_m\}$ the set of solutions to (s).

- $S \overset{?}{=} \mathcal{U} + P$

$P \to$ solution from S.

$\mathcal{U} \to$ vector subspace, described by (s).

$$D(S): \begin{cases} a_{11} \cdot x_1 + \dots + a_{1m} \cdot x_m = 0 \\ \vdots \\ a_{t1} \cdot x_1 + \dots + a_{tm} \cdot x_m = 0 \end{cases}$$

- $S \overset{?}{=} T$

$\hookrightarrow S \overset{?}{\subseteq} T$

Take $Q(q_1, \dots, q_m) \in S$.

$$\boxed{Q \in T \iff \overrightarrow{PQ} \in \mathcal{U}}$$

But:

$$a_{j1}(q_1 - p_1) + \dots + a_{jm} \cdot (q_m - p_m) =$$

$$= \underbrace{a_{j1} \cdot q_1 + \dots + a_{jm} \cdot q_m}_{b_j} - \underbrace{(a_{j1} \cdot p_1 + \dots + a_{jm} \cdot p_m)}_{b_j} =$$

$$\overset{?}{=} b_j - b_j$$

$$= 0$$

$? \Rightarrow \overrightarrow{PQ} \in \mathcal{U}$.

"$\Longleftarrow$" $\hookrightarrow T \subseteq S$, $(\forall) Q \in T \Rightarrow Q \in S$.

CHAPTER 4.

Proposition 4.14.

Volume of a parallelipiped P spanned by a, b, c is:

$$Vol(P) = |\langle a \times b, c \rangle| = \left| \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \right|$$

$a(a_1, a_2, a_3)$, $b(b_1, b_2, b_3)$, $c(c_1, c_2, c_3)$

→ components are w.r. to an orthonormal basis.

Proof.

- The area of the face spanned by a and b is: $|a \times b|$
- The height corresponding to this face: the length of the projection of c onto $a \times b$.

$$Vol(P) = \overset{A}{|a \times b|} \cdot \overset{h}{|pr^{\perp}_{a \times b}(c)|}$$

$$= |a \times b| \cdot \left| \frac{\langle a \times b, c \rangle}{\langle a \times b, a \times b \rangle} a \times b \right|$$

$$= |\langle a \times b, c \rangle|.$$

- W.r. to an orthonormal basis:

$$\langle a \times b, c \rangle = \left\langle \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}, c_1 \cdot i + c_2 \cdot j + c_3 \cdot k \right\rangle$$

$$= \begin{vmatrix} c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$